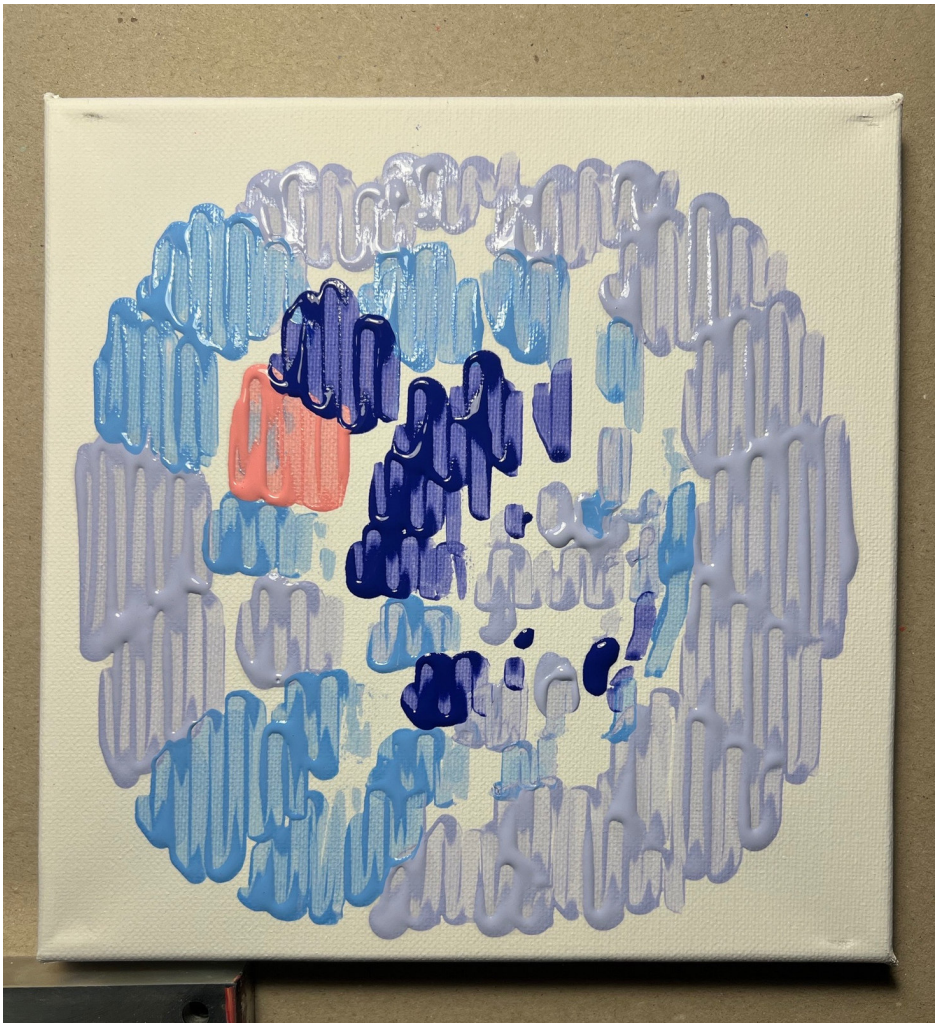
FIRST RESULTS / MACHINE PAINTING

The machine learns to paint

A first canvas, a new rubber-tipped tool, and the idea behind OPM-G: turning the machine's physical behavior into a distinct artistic language.



FIRST CANVAS / OPM-N / 20 × 20 CM

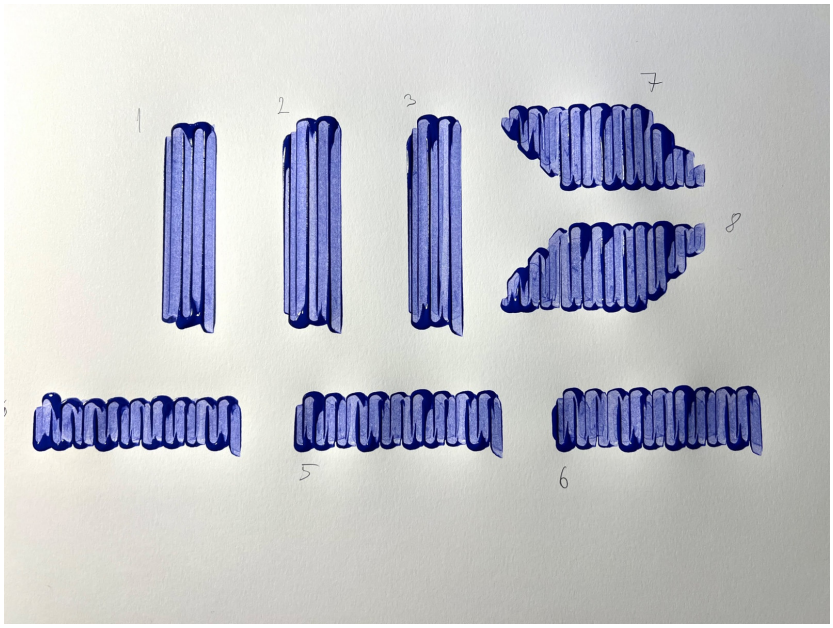
Key finding: the machine can already produce a recognizable stroke of its own. The next step is not to force it to copy computer graphics, but to design compositions from what that stroke can physically do.

01 / THE CENTRAL DISCOVERY

A visual language of its own

In the **IMG_2363.jpg** test, the rubber tip does more than deposit a band of color. Its marks combine three repeatable characteristics:

- **Geometry.** Almost perfectly straight, parallel lines.
- **Relief.** Two saturated edges with a thin film of paint between them.
- **Rhythm.** Distinct turns, rounded ends, and patterns of entry and exit.



Physical testing of the new tip: short, long, vertical and horizontal two-pass marks.

This is no longer an imitation of a human brush. Together, the geometry, relief, and rhythm create a machine-specific mark.

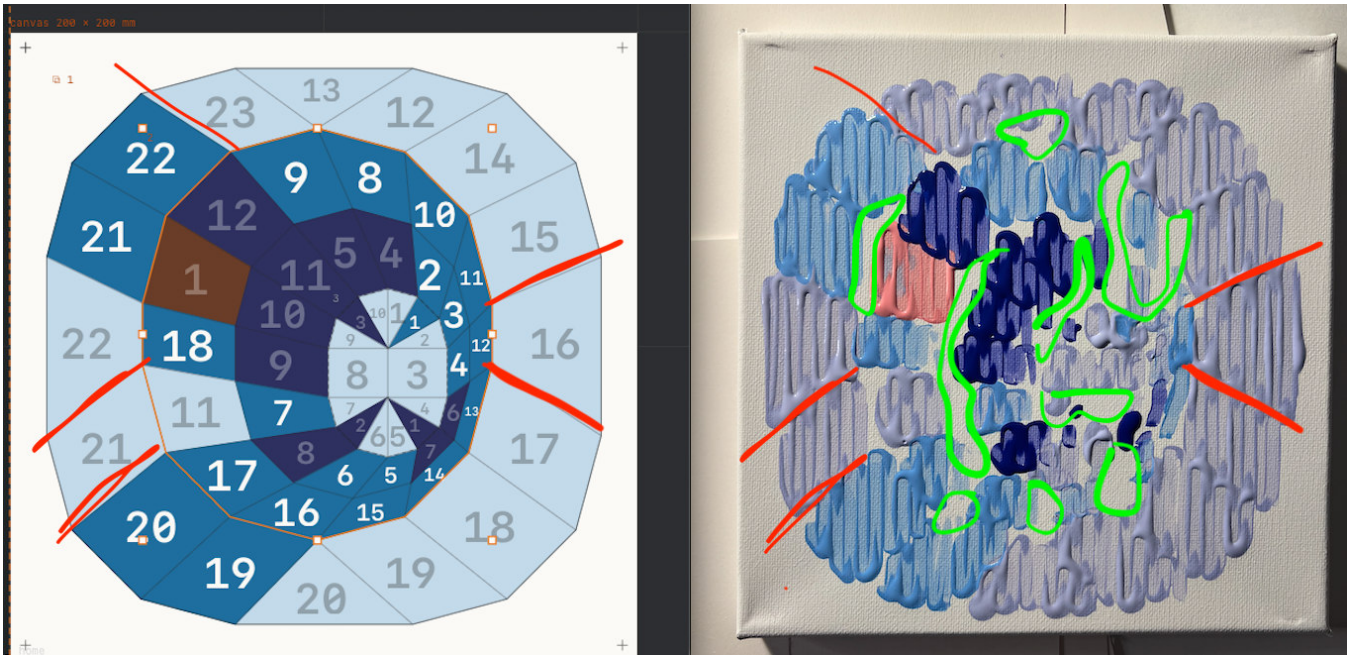
In the November 5 LAB tests, the middle sections of some stroke edges were estimated to deviate from a straight line by only hundredths of a millimeter. These were image-based estimates, not metrology, but the repeatability itself is a meaningful result.

> **THE DISCOVERY**
>
> The result is more than a tool for applying paint: it is a **repeatable physical graphic primitive** — the foundation of OPM-G.

02 / WHERE PLAN AND PAINT DIVERGE

The white gaps have several causes

Claude found that OPM-N fills cells independently: the outer strokes are inset from each cell boundary. Across a composition, adjacent insets leave unpainted gaps. Small cells and inconsistent paint supply make this worse.



Left: digital cell geometry. Right: the painted canvas, with red and green annotations indicating gaps and favored contours.

Claude proposes replacing many local grids with **one shared grid across the canvas**, then splitting its vertical lines by color. I agree: the image should use a single coordinate system rather than rebuilding one inside every shape.

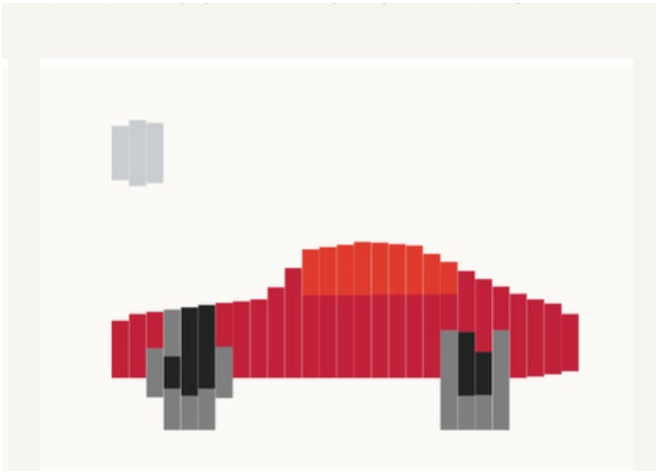
> **A CRITICAL TECHNICAL DETAIL**
>
> **idleplan.js** uses 5.83 mm and 4.38 mm spacing, an approximately 4 mm trace width, and a 3 mm minimum leg. In **strokes.js**, the entry/exit motion uses **SWEEP_TAIL = 8 mm**. A short digital segment is not necessarily a complete physical stroke.

The suggested 10 mm cutoff is therefore unproven. A short move may have no region of stable contact at all. We should measure the true minimum stroke length before automatically assigning tiny fragments to neighboring color areas.

03 / A DIFFERENT CONSTRUCTION METHOD

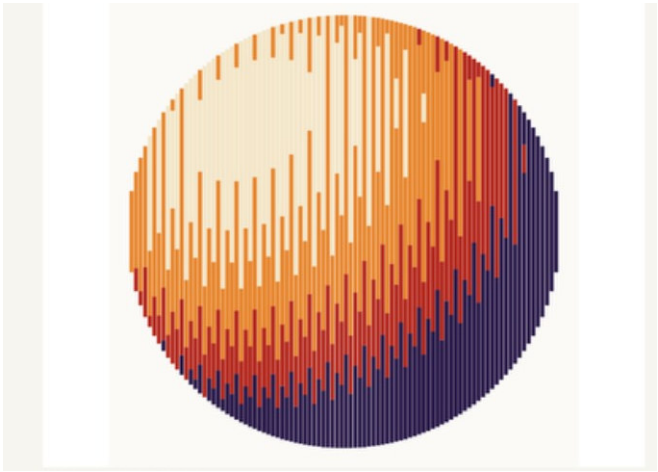
An image made from strokes

OPM-G should offer two artistic modes while sharing one generator of physical strokes.



GRAPHIC / FIGURATIVE

Cars, watches, human figures and signs. The priorities are legible silhouettes, contrast, and identifying details. At low resolution, a car becomes a pictogram; more columns restore recognizable form.



OPTICAL / COLOR MIXING

Color fields, gradients, and volume. Neighboring hues can interlock as teeth. The image emerges from many vertical strokes of different lengths and pigments.

The orange-red sphere is especially revealing. Its circle is not drawn as a curve. It emerges from the lengths and colors of straight marks planned for this specific tool.

Up close, a viewer sees paint relief and individual marks; farther away, a coherent form and mixed color may emerge. The actual viewing distance has not been measured. It depends on contrast, light, line spacing, and eyesight.

> **ARTISTIC POTENTIAL**
>
> The image may change not only with illumination but also as the viewer moves. The machine's straight, repeatable movements should be **revealed, not concealed**. They are its visual language.

04 / BEFORE BUILDING THE NEW TAB

Three physical tests

Start with one small test sheet and a minimal OPM-G geometry prototype, not a complete new interface. All numbers below are hypotheses to validate on real paint.

A MINIMUM STROKE LENGTH

Test **8, 12, 16, 24, and 32 mm** segments under identical tool settings. Examine the stable painted section, paint buildup at both ends, and whether short details remain controllable.

B GAP-FREE COVERAGE

Paint three adjacent rectangles at **5.83, 5.0, and 4.4 mm** pitch. Find the spacing that gives even two-pass coverage with no bare canvas while preserving the raised edges.

C TWO COLORS MEETING

Let a blue vertical stroke end where a lavender one begins. Compare painting against a **wet** neighbor with painting after the first color **dries**; also test neighboring parallel lines.

TEST SUBSTRATE

Use primed canvas, the same acrylic/pouring-medium mixture, and the same rubber tip. Paper may behave differently. Document each test before **and** after drying.

Then compare three views of the same painting in Rembrandt: **ideal geometry**, **predicted physical paint coverage**, and a **photo of the completed result**.

05 / THE NEXT STEP

Preserve what already works

Ask Claude to fix the edge-stroke bug in OPM-N separately, with regression tests protecting the verified LAB 6.0 behavior. Then calibrate the physical stroke and develop an OPM-G generator based on one shared grid.

There is no need to perfect OPM-N indefinitely. It has an aesthetic of its own: individual cells, each with its own pattern of paint. OPM-G is a different language: one continuous grid through which the image is constructed.



WHAT MUST NOT BE LOST

The first canvas still has accidental fragments and visible white gaps. Yet it already has something screen graphics cannot reproduce: **glossy, raised edges and a delicate, translucent center** within each stroke.

If the fill becomes completely uniform, we may erase the physical signature that made building the machine worthwhile in the first place.

A tinted ground would not merely hide gaps. The space between painted lines could become an intentional **third color** in the palette.

> REMBRANDT PRINCIPLE
>
> **Do not force the brush to reproduce an image. Build the image from what the brush can physically do.**

The machine already makes a recognizable mark. The next task for the artist and the software is to compose with it. This is a step beyond a geometric editor toward a model of the actual painting process.

Sources: first OPM-N canvas, IMG_2363 test, OPM-G sketches (Nov. 6, 2026). Code and research notes: github.com/forma78/Rembrandt (idleplan.js, strokes.js, OPM-N/README.md). Image-based estimates require physical validation.